

Markov-chains

DEF: Let (Ω, A, P) be a probability space, T is a time space. Then $X(t, \omega) : T * \Omega \rightarrow I$ a stochastic process iff $\forall t \in T: X(t, \omega) : \Omega \rightarrow I$ random variable.

DEF: $X_t(\omega) = X(t, \omega)$ stochastic process is a Markov-chain iff:
 $t_1 > t_0 \in T$ $A_{t_0} = \sigma(\{X_t\}_{t \leq t_0})$, $B \in B(I)$

$$\Rightarrow P(X_{t_1} \in B | A_{t_0}) = P(X_{t_1} \in B | X_{t_0})$$

Statement: Let $(X_n)_n$ be a discrete Markov-chain, μ is the probability distribution of X_0 (initial random variable) $\Rightarrow \mu$ and p_{ij} determines the Markov-chain.

Proof: $P(X_n = i_n, X_{n-1} = i_{n-1}, \dots, X_0 = i_0) = P(X_n = i_n | X_{n-1} = i_{n-1}, \dots, X_0 = i_0) \cdot P(X_{n-1} = i_{n-1}, \dots, X_0 = i_0) = P(X_n = i_n | X_{n-1} = i_{n-1}, \dots, X_0 = i_0) \cdot P(X_{n-1} = i_{n-1} | X_{n-2} = i_{n-2}, \dots, X_0 = i_0) \cdot P(X_{n-2} = i_{n-2}, \dots, X_0 = i_0) = P(X_n = i_n | X_{n-1} = i_{n-1}, \dots, X_0 = i_0) \cdot P(X_{n-1} = i_{n-1} | X_{n-2} = i_{n-2}, \dots, X_0 = i_0) \cdot \dots \cdot P(X_0 = i_0) = p_{i_n i_{n-1}} \cdot p_{i_{n-1} i_{n-2}} \cdot \dots \cdot p_{i_0 i_1} \cdot \mu_{i_0}$. $\square$

Statement(Chapman-Kolmogorov equality): Let $(X_n)_n$ be a discrete Markov-chain. Then

$$p_{ij}(n+m) = \sum_{k \in I} p_{ik}(n) \cdot p_{kj}(m)$$

Proof: $P(X_{n+m} = j, X_n = k) = P(X_{n+m} = j | X_n = k) \cdot P(X_n = k)$.
 $P(X_{n+m} = j | X_0 = i) = \sum_{k \in I} P(X_{n+m} = j, X_n = k | X_0 = i) = P(X_{n+m} = j | X_n = k, X_0 = i) \cdot P(X_n = k | X_0 = i)$ $\square$

Corollary: The n -step transition probability matrix is P^n , where P is the 1-step transition probability matrix.

Proof: Using induction: For $n = 1$ is trivial. For $n = 2$ is true because of the formula of matrix multiplication. For $n > 2$ is the same (formula of matrix multiplication) $\square$